SOP Number and version: SOP-NAT-THIGINF.v01

SOP Title: **Mouse Thigh Infection Model**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
Author	Research name	Student		01/10/2024
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Approval	Frank J. Hernandez	Principal investigator		09/01/2025

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1. Overview and background

Mouse thigh infection model (pyomyositis) is one of the most used animal models in antibiotic development. The model can be applied in immunosuppressed or immunocompetent mice. In Nucleic Acids Technology laboratories, immunocompetent mice are used for this purpose unless immunosuppression is necessary.

2. Scope

In this SOP, *in-vivo* mouse thigh infection model for testing therapeutic efficacy of a drug substance in Nucleic Acids Technology laboratories is explained. All animal experiments are performed in Centrum för Biomedicinska Resurser (CBR) located at Campus US of Linköping University with the ethical approval by the Swedish Board of Agriculture under the **animal protocol Dnr 1253 (2018-2023) and Dnr 14980-2023 (2023-2028).** All experiments are conducted according to the principles expressed in the Declaration of Helsinki. Preparation of bacteria and analysis of the results are performed in Nucleic Acids Technology laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686).

This SOP covers the sections below:

- Bacteria preparation
- Infection induction
- Treatment
- Analysis

3. Procedure

Before starting the experiment, design entire experimental plan including the properties (e.g., strain, sex, age, number) of the mice, experimental groups (e.g., placebo, SoC, and test drug), timeline (starting from preparation) and dosing regimen.
This experimental plan needs to be uploaded to the CBR database (tick@lab) and approved by the Principal Investigator (PI) and CSO, Frank Hernandez, and CBR before starting.
Also, make sure that all needed instruments (e.g., trimmer, heating lamp, restrainer, injection lamps with magnifier, anesthesia equipment, surgical equipment, bead sterilizer, tissue homogenizer, incubator, colony counter) and disposables (e.g., drug substances, injectors, anesthetics, PPE, homogenization tubes, media) are ready to use at least the day before starting.
Any delay/change in the timeline/plan needs to be reported first to the PI (Frank Hernandez), and then to CBR.

3.1. Bacteria preparation

- Cultivate the strain that will be used for the infection model from the storage vials as described in **SOP-NAT-CULT**.
- Use a single colony from a pure culture to make a 25-50 mL liquid culture as described in **SOP-NAT-CULT**.
- Incubate the liquid culture at 36±1°C at 100-120 rpm shaking for 12 hours.
- Measure the McFarland turbidity of the liquid culture and adjust the bacteria concentration to 10^7 CFU/mL. This can be adjusted based on the requirements of different pathogens to cause infection in desired severity, if needed.
- Perform the control cultures for confirming the calculated number of bacteria as described in **SOP-NAT-CFU**.

3.2. Infection induction

The aim is to induce reproducible infections with desired severity which is the level that causes a visible localized infection with moderate severity (e.g., no large area of tissue necrosis) **(Figure 1).**

- Shave the mice under 2% isoflurane anesthesia
- Disinfect the right thigh area using wipes with alcohol-based disinfectant
- Inject 50 µL prepared bacteria solution into right thigh muscle
- Make sure that all the volume remains in the muscle after the injection
- Disinfect the leg one more time before transferring the mice back to the dedicated cage

Figure 1. Visible infection with moderate severity.

3.3. Treatment

- Inject desired concentration/volume of the drug substance (e.g., test drug, control drug, placebo) 2 hours after infection induction (e.g., 100 µL i.v. of 1 mg/mL)
- If needed, continue to inject further doses until 24 hours (e.g., q6h)
- Euthanize all the mice 24 hours after the infection induction by cervical decapitation under 2% isoflurane anesthesia.
- If needed, collect blood samples into recommended tubes (e.g., tubes with EDTA for blood counts) by cardiac puncture under ketamine/dexamethasone anesthesia (instead of isoflurane) before euthanasia.

3.3.1. Blood collection

- For blood collection, anesthetize the mice by intraperitoneal injection 75 mg/kg ketamine and 1 mg/kg of dexmedetomidine mixture in 100 µL for 10-gram body weight.
- Confirm the efficacy of the anesthesia by the loss of righting reflex of the mice.
- Place the mice dorsally on the surgery pad.
- Insert a 29-gauge needle fitted to a 1 mL syringe slowly below the sternum.
- Once the needle reached the mouse heart and blood in the syringe, collect at least 250 µL of blood into collection tubes with EDTA for veterinary use **(Figure 2**).
- If needed, collect another at least 250 µL blood into microcentrifuge tubes for serum biochemistry analysis.

Figure 2. EDTA blood collection tubes for veterinary use.

3.3.2. Serum preparation

- Separate the serum by centrifugation at 3000 rpm for 10 minutes and store at +4 °C until analysis. Store the samples at -20 °C if the time before analysis is longer than 24 hours.

This may change based on the targeted parameters (contact the responsible external laboratory for further information, if needed)

3.3.3. Tissue collection

- Take pictures of the infected/uninfected legs to record the visual inspection results.
- Remove right (infected) and left (uninfected control) thigh muscles under sterile conditions.
- Put these samples into previously labelled microcentrifuge tubes.
- Transfer all the samples to Nucleic Acids Technology laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) in dedicated biological material transfer system **(Figure 3).**

Figure 3. UN3373 triple packaging infectious material transfer system.

3.4. Analysis

3.4.1. Blood analysis

- Use a calibrated veterinary hematology analyzer and biochemistry analyzer with quality control blood samples for blood count and serum biochemistry analysis.
- Report the results as described in **SOP-NAT-DATA**.

3.4.2. Microbiological analysis

- Weigh thigh muscle tissues using a high precision analytical lab scale.
- Transfer each sample into screwed capped homogenization tubes with ceramic beads **(Figure 4).**
- Add 1 mL of PBS.
- Place the tubes into bead homogenizer **(Figure 5).**
- Homogenize the samples at the highest speed for 5 minutes.
- Perform the serial dilutions of the samples for enumeration of bacteria as described in **SOP-NAT-CFU**.
- Inoculate each dilution onto agar plates and incubate at 36±1°C for 18-24 hours as described in **SOP-NAT-CFU**.

- Complete the enumeration by counting the bacterial colonies and calculating the CFU/thigh and CFU/gram as described in **SOP-NAT-CFU**.
- Calculate log reduction and % reduction in CFU compared to controls (placebo and comparison antibiotic if used) and initial inoculation (e.g., $5x10^5$ CFU).
- Report the results as described in **SOP-NAT-DATA**.

Figure 4. Homogenization tubes.

Figure 5. Automated tissue homogenizer

3.4.3. Histopathological analysis

For histopathological analysis, Nucleic Acids Technology laboratories collaborate with Microscopy Core Facility of Linköping University (Campus US, House 462, Entrance 71, Level 10).

- Transfer the muscle tissues into 4% buffered formaldehyde solution and store at 4°C for 24 hours.
- Remove the tissues from formaldehyde solution and transfer into 50% ethanol for 15 minutes followed by 70% ethanol, until processed.
- Fix, dehydrate and wax infiltrate the tissues using a Leica TP1020 system for 7 hours.
- Embed the tissues in paraffin using a Leica EG 1150 system.
- Prepare sections using an Automatic Rotary Microtome HM355S system.
- Stain the samples with Hematoxylin-Eosin (H&E) using an Automatic Leica ST4020 Linear Stainer.
- Image the slides using an Olympus BX51 light microscope with camera.
- Take pictures of the related areas and report as described in **SOP-NAT-DATA**.

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology laboratories staff on performing therapeutic efficacy analysis *in-vivo* mouse thigh infection model used in Nucleic Acids Technology laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer

- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- µL – Microliter
- mL – Milliliter
- CFU – Colony Forming Units
- PBS – Phosphate Buffered Saline
- CBR – Centrum för Biomedicinska Resurser
- PI – Principal Investigator
- EDTA – Ethylenediaminetetraacetic acid
- Rpm - Revolutions Per Minute

6. Responsibilities

This SOP was prepared by Student name, revised by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- High precision analytical balance
- BSL-2 biosafety cabinet
- Petri dishes
- Glass flasks
- Aluminum foil
- Inoculation loops
- Bunsen burner
- Gloves
- Pipettes
- Pipette tips
- Incubator
- Syringes
- Blood collection tubes with EDTA
- Microcentrifuge tubes
- Surgical instruments
- Bead sterilizer
- Bead sterilizer tubes with ceramic beads
- Camera and charger
- SD card
- Trimmer and charger
- Automatized tissue homogenizer
- Homogenization tubes
- Automatized colony counter
- UN3373 triple packaging infectious material transfer system

- Wet wiped with alcohol
- Heating lamp (provided by CBR)
- Lamp with magnifier (provided by CBR)
- Animal housing material (provided by CBR)
- Anesthesia equipment (provided by CBR)
- Histopathology equipment (provided by Microscopy Core Facility)

8. Caution/Warning

- Use BSL-2 biosafety cabinet for all the work that involves bacteria.
- Use PPE according to **SOP-NAT-BSL2**.
- Clean the working area before and after work.

9. References

- Borsa BA et. al. J Control Release, 2023; 361 (2023) 260-269.

10. Version Control

Original version of the **SOP-NAT-THIGINF.v01**. Check **Table 1**.

11.Attachments/Appendices

Appendix I:

- Bunsen burner manual
- Automated homogenizer manual
- Automated colony counter manual
- Lab scale manual
- BSL-2 biosafety cabinet manual
- Ethical approval